# BeamLib Theory Manual
# Chapter 1: Euler–Bernoulli 2D Beam Element

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Phase 1, Batch 2A

## 1 Scope and Assumptions

This chapter derives the 2-node planar Euler–Bernoulli (EB) beam element used in BeamLib for slender beams ($L/h > 20$). The element operates in the global $xz$-plane.

**Kinematic assumptions.**

1. Plane sections remain plane after deformation.
2. Cross-section normals remain normal to the deformed beam axis (no shear deformation; this is the defining EB assumption).
3. Displacements and rotations are infinitesimal (linear theory).
4. Material is linear elastic, isotropic, homogeneous within the section.

The element is restricted to in-plane bending about the $y$-axis with axial extension along the local $x$-axis. Torsion and out-of-plane bending are not modeled here (they are 3D effects, handled in EB3D).

## 2 Sign and DOF Conventions

### 2.1 Coordinate system

We use a right-handed Cartesian frame with axes $(x, y, z)$ satisfying $\hat{x} \times \hat{y} = \hat{z}$. The 2D problem lives in the $xz$-plane:

- $x$: in-plane axial reference direction.
- $z$: in-plane transverse direction.
- $y$: out-of-plane axis, oriented so that $(\hat{x}, \hat{y}, \hat{z})$ form a right-handed frame ($+\hat{y}$ points *out* of the page in the standard view with $+\hat{x}$ rightward and $+\hat{z}$ upward).

**Important.** Positive $\theta_y$ in BeamLib is defined by the structural-mechanics convention $\theta_y = \mathrm{d}w/\mathrm{d}x$ (see §2.4). This is *opposite* in sign to the strict right-hand-rule rotation about $+\hat{y}$. Throughout this manual "$\theta_y$" means the BeamLib structural convention unless explicitly noted.

## 2.2 Per-node DOFs

At every node $i$ we store three DOFs in the order

$$\boldsymbol{d}_i = \begin{bmatrix} u_{x,i} \\ u_{z,i} \\ \theta_{y,i} \end{bmatrix}.$$

Positive $u_x$ is in the $+x$ direction (global). Positive $u_z$ is in the $+z$ direction (global). Positive $\theta_y$ follows the BeamLib structural convention $\theta_y = \mathrm{d}w/\mathrm{d}x$ (see §2.4; opposite sign vs. right-hand rule about $+\hat{y}$).

## 2.3 Element DOF vector

A 2-node EB2D element has 6 DOFs ordered as

$$\boldsymbol{d}^e = \begin{bmatrix} \boldsymbol{d}_A \\ \boldsymbol{d}_B \end{bmatrix} = \begin{bmatrix} u_{xA}, & u_{zA}, & \theta_{yA}, & u_{xB}, & u_{zB}, & \theta_{yB} \end{bmatrix}^{\mathsf{T}}.$$

Indices in this chapter run $0, \ldots, 5$ to match the C++ implementation.

## 2.4 Sign convention table

| Quantity | Symbol | Positive direction |
|---|---|---|
| Axial displacement | $u_x$ | along $+\hat{x}$ (global $x$) |
| Transverse displacement | $u_z$ | along $+\hat{z}$ (global $z$) |
| Section rotation | $\theta_y$ | structural convention: $\theta_y = \mathrm{d}w/\mathrm{d}x$ (positive slope ⇒ positive |
| Axial force on a node | $F_x$ | along $+\hat{x}$ |
| Transverse force on a node | $F_z$ | along $+\hat{z}$ |
| Nodal moment | $M_y$ | work-conjugate to $\theta_y$ (same convention; positive $M_y$ produces |
| Slope-rotation relation | $\theta_y(x) = \mathrm{d}w/\mathrm{d}x$ | with $w \equiv u_z$ |

Table 1: Sign conventions for the EB2D element.

**Justification of $\theta_y = \mathrm{d}w/\mathrm{d}x$ (structural convention).** BeamLib adopts the standard structural-mechanics convention: a positive section rotation $\theta_y$ corresponds to a positive slope $\mathrm{d}w/\mathrm{d}x$. Geometrically, in the $xz$-plane as drawn ($+\hat{x}$ rightward, $+\hat{z}$ upward), this is a counter-clockwise rotation of the cross-section as the viewer reads the figure.

This is the *opposite* sign from a strict right-hand-rule rotation about $+\hat{y}$. By Rodrigues' formula,

$$R_y^{\mathrm{RH}}(\theta)\hat{x} = (\cos\theta,\, 0,\, -\sin\theta),$$

so a positive right-hand-rule rotation tilts the cross-section normal toward $-\hat{z}$, not $+\hat{z}$. The BeamLib convention is therefore equivalent to a right-hand-rule rotation about $-\hat{y}$.

We adopt the structural convention because (i) it makes bending kinematics intuitive: positive slope ⇒ positive rotation, no mental flip needed when reading a diagram; (ii) it matches the standard Hermite shape function $H_2(\xi) = L(\xi - 2\xi^2 + \xi^3)$, whose derivative at $\xi = 0$ is $+1$, enforcing $\theta_{yA} = +\mathrm{d}u_z/\mathrm{d}x$ at node $A$; (iii) it matches the MATLAB `JaphyBeam` reference and most structural-mechanics textbooks. The convention propagates consistently to every $k_l$ sign in §5, the residual definition $\boldsymbol{r}_l = \boldsymbol{k}_l\,\boldsymbol{d}_l$, and the verification signs in §10.

**Cross-check.** For a cantilever with downward tip load $F_z < 0$: the beam droops ($u_z(L) < 0$), the slope is negative at the tip ($\mathrm{d}w/\mathrm{d}x < 0$), so under the BeamLib convention $\theta_y(L) = \mathrm{d}w/\mathrm{d}x < 0$. The test `test_eb2d_patch.cpp` asserts this sign explicitly.

# 3 Strain–Displacement Relations

Let $u(x,z)$, $w(x,z)$ denote total displacement of a material point. EB kinematics:

$$u(x,z) = u_x(x) - z\,\theta_y(x) = u_x(x) - z\,\frac{\mathrm{d}w}{\mathrm{d}x}, \qquad w(x,z) = u_z(x).$$

The axial strain is

$$\varepsilon_{xx} = \frac{\partial u}{\partial x} = \frac{\mathrm{d}u_x}{\mathrm{d}x} - z\,\frac{\mathrm{d}^2 w}{\mathrm{d}x^2}.$$

The bending curvature is $\kappa = \mathrm{d}^2 w/\mathrm{d}x^2 = \mathrm{d}^2 u_z/\mathrm{d}x^2$.

The strain energy density per unit length, after integrating over the cross-section, is

$$\frac{1}{2}\int_A E\,\varepsilon_{xx}^2\,\mathrm{d}A = \frac{1}{2}\,EA\left(\frac{\mathrm{d}u_x}{\mathrm{d}x}\right)^2 + \frac{1}{2}\,EI_z\left(\frac{\mathrm{d}^2 u_z}{\mathrm{d}x^2}\right)^2,$$

where $I_z = \int_A z^2\,\mathrm{d}A$ is the second moment of area about the $y$-axis (the axis perpendicular to the bending plane, i.e. $\hat{y}$). **In BeamLib we store this as** `SectionProperties::Iz`, consistent with the 2D xz-plane bending convention.

**Section property used.** 2D EB bending in the $xz$-plane uses `props.Iz`. The 2D–3D cross-validation table in PROJECT_SPEC §13.6 maps 2D `Iz` ↔ 3D `Iy` when comparing the same physical scenario.

# 4 Shape Functions

Use a 2-node element on the natural coordinate $\xi = x/L \in [0,1]$.

## 4.1 Axial (linear)

$$N_1^a(\xi) = 1 - \xi, \qquad N_2^a(\xi) = \xi.$$

Axial displacement:

$$u_x(\xi) = N_1^a(\xi)\,u_{xA} + N_2^a(\xi)\,u_{xB}.$$

## 4.2 Transverse (Hermite cubic)

Four cubic Hermite functions interpolate $u_z$ and $\theta_y = \mathrm{d}u_z/\mathrm{d}x$ at the two ends:

$$\begin{aligned} H_1(\xi) &= 1 - 3\xi^2 + 2\xi^3, & H_2(\xi) &= L\,(\xi - 2\xi^2 + \xi^3),\\ H_3(\xi) &= 3\xi^2 - 2\xi^3, & H_4(\xi) &= L\,(-\xi^2 + \xi^3). \end{aligned}$$

Transverse displacement:

$$u_z(\xi) = H_1\,u_{zA} + H_2\,\theta_{yA} + H_3\,u_{zB} + H_4\,\theta_{yB}.$$

Verification at the endpoints (using $\theta_y = \mathrm{d}u_z/\mathrm{d}x = (1/L)\,\mathrm{d}u_z/\mathrm{d}\xi$):

$$u_z(0) = u_{zA}, \quad u_z(1) = u_{zB}, \quad \left.\frac{\mathrm{d}u_z}{\mathrm{d}x}\right|_{\xi=0} = \theta_{yA}, \quad \left.\frac{\mathrm{d}u_z}{\mathrm{d}x}\right|_{\xi=1} = \theta_{yB}.$$

### 4.3 Curvature operator

Let $\boldsymbol{d}_b = (u_{zA}, \theta_{yA}, u_{zB}, \theta_{yB})^\mathsf{T}$. Then

$$\kappa(\xi) = \frac{\mathrm{d}^2 u_z}{\mathrm{d}x^2} = \boldsymbol{B}_b(\xi)\, \boldsymbol{d}_b,$$

with

$$\boldsymbol{B}_b(\xi) = \frac{1}{L^2}\begin{bmatrix} -6+12\xi & L(-4+6\xi) & 6-12\xi & L(-2+6\xi) \end{bmatrix}.$$

## 5 Element Stiffness Matrix Derivation

### 5.1 Axial stiffness

$$\boldsymbol{k}^a_{2\times 2} = \int_0^L EA \begin{bmatrix} N_1'^2 & N_1' N_2' \\ N_1' N_2' & N_2'^2 \end{bmatrix} \mathrm{d}x = \frac{EA}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

### 5.2 Bending stiffness

$$\boldsymbol{k}^b_{4\times 4} = \int_0^L EI_z\, \boldsymbol{B}_b^\mathsf{T} \boldsymbol{B}_b \,\mathrm{d}x.$$

Performing the integral yields the well-known result

$$\boldsymbol{k}^b = \frac{EI_z}{L^3}\begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}.$$

### 5.3 Assembled element stiffness $\boldsymbol{k}_l$ in DOF order $[u_{xA}, u_{zA}, \theta_{yA}, u_{xB}, u_{zB}, \theta_{yB}]$

Mapping the axial block to indices $(0, 3)$ and the bending block to indices $(1, 2, 4, 5)$, the nonzero entries of the local $6 \times 6$ stiffness $\boldsymbol{k}_l$ are:

$$\begin{aligned}
&k_l(0,0) = k_l(3,3) = \tfrac{EA}{L}, && k_l(0,3) = k_l(3,0) = -\tfrac{EA}{L},\\
&k_l(1,1) = k_l(4,4) = \tfrac{12EI_z}{L^3}, && k_l(1,4) = k_l(4,1) = -\tfrac{12EI_z}{L^3},\\
&k_l(1,2) = k_l(2,1) = \tfrac{6EI_z}{L^2}, && k_l(1,5) = k_l(5,1) = \tfrac{6EI_z}{L^2},\\
&k_l(2,2) = k_l(5,5) = \tfrac{4EI_z}{L}, && k_l(2,5) = k_l(5,2) = \tfrac{2EI_z}{L},\\
&k_l(2,4) = k_l(4,2) = -\tfrac{6EI_z}{L^2}, && k_l(4,5) = k_l(5,4) = -\tfrac{6EI_z}{L^2}.
\end{aligned}$$

All other entries are zero. The matrix is symmetric.

### 5.4 Residual convention

For this linear element the residual (internal force vector) is

$$\boxed{\boldsymbol{r}^e_l = \boldsymbol{k}_l\, \boldsymbol{d}^e_l.}$$

The Newton residual at the global level is $\boldsymbol{R} = \boldsymbol{R}_{\text{int}} - \boldsymbol{F}_{\text{ext}}$ and the correction equation is $\boldsymbol{K}\,\delta = -\boldsymbol{R}$. With this sign, the convergent state satisfies $\boldsymbol{R}_{\text{int}} = \boldsymbol{F}_{\text{ext}}$.

# 6 Consistent Mass Matrix

The consistent mass matrix for EB2D uses the same shape functions:

$$\boldsymbol{m}^a_{2\times 2} = \int_0^L \rho A \begin{bmatrix} N_1^a N_1^a & N_1^a N_2^a \\ N_1^a N_2^a & N_2^a N_2^a \end{bmatrix} \mathrm{d}x = \frac{\rho A L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

$$\boldsymbol{m}^b_{4\times 4} = \int_0^L \rho A\, \boldsymbol{H}^\mathsf{T} \boldsymbol{H}\, \mathrm{d}x = \frac{\rho A L}{420} \begin{bmatrix} 156 & 22L & 54 & -13L \\ 22L & 4L^2 & 13L & -3L^2 \\ 54 & 13L & 156 & -22L \\ -13L & -3L^2 & -22L & 4L^2 \end{bmatrix}.$$

EB2D mass does *not* include rotary inertia (this is the standard textbook consistent mass for EB and matches PROJECT_SPEC §3.4). Embedded in the $6 \times 6$ element DOF order:

$$\begin{aligned}
&m_l(0,0) = m_l(3,3) = \tfrac{\rho A L}{3}, && m_l(0,3) = m_l(3,0) = \tfrac{\rho A L}{6}, \\
&m_l(1,1) = m_l(4,4) = \tfrac{156\,\rho A L}{420}, && m_l(1,4) = m_l(4,1) = \tfrac{54\,\rho A L}{420}, \\
&m_l(1,2) = m_l(2,1) = \tfrac{22L\,\rho A L}{420}, && m_l(1,5) = m_l(5,1) = -\tfrac{13L\,\rho A L}{420}, \\
&m_l(2,2) = m_l(5,5) = \tfrac{4L^2\,\rho A L}{420}, && m_l(2,5) = m_l(5,2) = -\tfrac{3L^2\,\rho A L}{420}, \\
&m_l(2,4) = m_l(4,2) = \tfrac{13L\,\rho A L}{420}, && m_l(4,5) = m_l(5,4) = -\tfrac{22L\,\rho A L}{420}.
\end{aligned}$$

This matrix is implemented in Batch 2A as part of `computeMass`, but the assembly path `BeamModel::assembleMass` is left for Batch 2B.

# 7 Coordinate Transformation

Each element is integrated in a canonical *local* frame where node $A$ sits at the origin and node $B$ at $(L, 0, 0)$. The transformation $\boldsymbol{T}$ relates global element DOFs to local DOFs.

## 7.1 Per-node block

Let $\Delta\boldsymbol{x} = \boldsymbol{x}_B - \boldsymbol{x}_A$ in global coordinates with components $\Delta x_x$ and $\Delta x_z$ (the $y$ component is zero in the 2D problem). Define

$$L = \|\Delta\boldsymbol{x}\|, \qquad c = \frac{\Delta x_x}{L}, \qquad s = \frac{\Delta x_z}{L}.$$

The $3 \times 3$ per-node block is

$$\boldsymbol{T}_n = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The element-level $6 \times 6$ matrix is block-diagonal: $\boldsymbol{T} = \mathrm{diag}(\boldsymbol{T}_n, \boldsymbol{T}_n)$.

## 7.2 Application convention

By construction $\boldsymbol{T}$ is orthogonal ($\boldsymbol{T}^\mathsf{T}\boldsymbol{T} = \boldsymbol{I}$). BeamLib applies the transformation as

$$\boldsymbol{d}_l = \boldsymbol{T}\,\boldsymbol{d}_g, \qquad \boldsymbol{r}_g = \boldsymbol{T}^\mathsf{T}\boldsymbol{r}_l, \qquad \boldsymbol{k}_g = \boldsymbol{T}^\mathsf{T}\boldsymbol{k}_l\,\boldsymbol{T}.$$

This convention matches the MATLAB reference (`JaphyBeam/JaphyEulerBernoulliBeam2D/calcTransformat` and the user spec verbatim.

**Sanity checks.**

- Horizontal beam ($\boldsymbol{x}_B - \boldsymbol{x}_A = (L, 0, 0)$): $c = 1$, $s = 0$, $\boldsymbol{T}_n = \boldsymbol{I}_3$, so local and global frames coincide.
- Orthogonality: $\boldsymbol{T}_n^\mathsf{T}\boldsymbol{T}_n = \mathrm{diag}(c^2 + s^2,\, c^2 + s^2,\, 1) = \boldsymbol{I}_3$ since $c^2 + s^2 = 1$.
- Energy invariance: $\boldsymbol{d}_g^\mathsf{T}\boldsymbol{k}_g\boldsymbol{d}_g = (\boldsymbol{T}\boldsymbol{d}_g)^\mathsf{T}\boldsymbol{k}_l(\boldsymbol{T}\boldsymbol{d}_g) = \boldsymbol{d}_l^\mathsf{T}\boldsymbol{k}_l\boldsymbol{d}_l$.
- The $\theta_y$ DOF is invariant ($T_n(2,2) = 1$) because the rotation axis $\hat{y}$ is preserved by 2D in-plane rotation about $\hat{y}$.

**Note on Batch 2A scope.** All Batch 2A tests use beams aligned with the global $x$-axis, so $\boldsymbol{T} = \boldsymbol{I}$ and the transformation is the identity. The non-trivial rotated-beam verification is deferred to Batch 2B (PROJECT_SPEC §2B.5).

### 7.3 Reference vector

`computeTransformation(xA, xB, refVector)` takes a `refVector` argument for API uniformity with 3D elements but **ignores it** in 2D. The 2D rotation about $+\hat{y}$ is fully determined by the in-plane direction $(\Delta x_x, \Delta x_z)$.

## 8 Global Assembly

For each element with connectivity $(A, B)$:

1. Build local $\boldsymbol{T}$ from $\boldsymbol{x}_A$, $\boldsymbol{x}_B$.
2. Gather global element DOFs $\boldsymbol{d}_g^e$ from nodal storage.
3. $\boldsymbol{d}_l^e = \boldsymbol{T}\,\boldsymbol{d}_g^e$.
4. Pass $\boldsymbol{x}_A^{loc} = (0, 0, 0)$, $\boldsymbol{x}_B^{loc} = (L, 0, 0)$, $\boldsymbol{d}_l^e$ to `EulerBernoulli2D::computeElement` to obtain $\boldsymbol{r}_l^e$ and $\boldsymbol{k}_l^e$.
5. $\boldsymbol{r}_g^e = \boldsymbol{T}^\mathsf{T}\boldsymbol{r}_l^e$, $\boldsymbol{k}_g^e = \boldsymbol{T}^\mathsf{T}\boldsymbol{k}_l^e\boldsymbol{T}$.
6. Scatter $\boldsymbol{r}_g^e$, $\boldsymbol{k}_g^e$ into the free-DOF system using the DofMap. Entries with any fixed DOF on either side are dropped (Batch 2A only supports homogeneous Dirichlet constraints; PROJECT_SPEC §11).

## 9 Formula-to-C++ Index Mapping

| Formula symbol | C++ index | Meaning |
|---|---|---|
| $u_{xA}$ | `dispVec[0]` | node-A axial displacement (global $x$ in canonical local frame) |
| $u_{zA}$ | `dispVec[1]` | node-A transverse displacement |
| $\theta_{yA}$ | `dispVec[2]` | node-A section rotation |
| $u_{xB}$ | `dispVec[3]` | node-B axial displacement |
| $u_{zB}$ | `dispVec[4]` | node-B transverse displacement |
| $\theta_{yB}$ | `dispVec[5]` | node-B section rotation |
| $k_l(0,0)$ | `ke(0,0)` | $EA/L$ |
| $k_l(3,3)$ | `ke(3,3)` | $EA/L$ |

| Formula symbol | C++ index | Meaning |
|---|---|---|
| $k_l(0,3)$ | `ke(0,3)` | $-EA/L$ |
| $k_l(1,1)$ | `ke(1,1)` | $12EI_z/L^3$ |
| $k_l(1,2)$ | `ke(1,2)` | $6EI_z/L^2$ |
| $k_l(1,4)$ | `ke(1,4)` | $-12EI_z/L^3$ |
| $k_l(1,5)$ | `ke(1,5)` | $6EI_z/L^2$ |
| $k_l(2,2)$ | `ke(2,2)` | $4EI_z/L$ |
| $k_l(2,4)$ | `ke(2,4)` | $-6EI_z/L^2$ |
| $k_l(2,5)$ | `ke(2,5)` | $2EI_z/L$ |
| $k_l(4,4)$ | `ke(4,4)` | $12EI_z/L^3$ |
| $k_l(4,5)$ | `ke(4,5)` | $-6EI_z/L^2$ |
| $k_l(5,5)$ | `ke(5,5)` | $4EI_z/L$ |
| $c$ | `c` | $\Delta x_x/L$ |
| $s$ | `s` | $\Delta x_z/L$ (note: $z$, not $y$, in the xz-plane) |
| $T_n(0,1)$ | `T(0,1)` | $-s$ |
| $T_n(1,0)$ | `T(1,0)` | $+s$ |

Table 2: Formula symbol to C++ index mapping for EB2D.

# 10 Verification: Analytical Solutions

## 10.1 Cantilever tip transverse load

A cantilever of length $L$ fixed at node 0 with a transverse load $F_z$ at the tip satisfies

$$u_z(L) = \frac{F_z\, L^3}{3EI_z}, \qquad \theta_y(L) = \frac{F_z\, L^2}{2EI_z}.$$

For $F_z < 0$ (downward), both are negative (tip droops, slope is negative). With $F_z = -F$ ($F > 0$ being the magnitude),

$$u_z(L) = -\frac{FL^3}{3EI_z}, \qquad \theta_y(L) = -\frac{FL^2}{2EI_z}.$$

Reference numerical values used in `test_eb2d_cantilever.cpp`: $E = 200 \times 10^9$, $A = 0.01$, $I_z = 8.333 \times 10^{-6}$, $L = 1.0$, $F = 1000$, giving $u_z(L) = -2 \times 10^{-4}$ and $\theta_y(L) = -3 \times 10^{-4}$.

## 10.2 Cantilever tip end-moment

A cantilever with an applied end moment $M_y$ at the tip yields a constant curvature $\kappa = M_y/(EI_z)$ throughout the beam and

$$u_z(L) = \frac{M_y\, L^2}{2EI_z}, \qquad \theta_y(L) = \frac{M_y\, L}{EI_z}.$$

The bending moment is constant along the span (no shear, pure bending).

## 10.3 Patch test (constant axial strain)

Apply $u_{xB} = u_0$ at the right end of an arbitrary mesh of EB2D elements aligned with $+\hat{x}$, with the left end pinned axially. Every element should report axial strain $u_0/L_{\text{total}}$ and zero bending response.

### 10.4 Rigid translation

Setting $u_{xA} = u_{xB}$, $u_{zA} = u_{zB}$, $\theta_{yA} = \theta_{yB} = 0$ should give $\boldsymbol{r}_l = \boldsymbol{k}_l \boldsymbol{d}_l$ where the axial block contributes zero (equal axial DOFs) and bending block also gives zero (constant deflection, zero curvature, zero rotation difference). Hence $\boldsymbol{r}_l = \boldsymbol{0}$.

## 11 Verification Tolerance Table

| Check | Tolerance |
|---|---|
| Single-element matrix entries ($EA/L$, $12EI/L^3$, ...) | relative $10^{-12}$ |
| Patch test rigid-translation residual | absolute $10^{-12}$ |
| Patch test constant-strain residual | absolute $10^{-10}$ |
| Cantilever tip $u_z$, $\theta_y$ (10 elements) | relative $10^{-9}$ |
| Pure end-moment $u_z$, $\theta_y$ (10 elements) | relative $10^{-9}$ |
| Constant-moment uniformity along span | relative $10^{-9}$ |
| Newton iteration count (linear case) | exactly 1 correction step |
| Final NR residual (linear case) | $\leq 10^{-7}$ absolute (set by NRConfig.tol) |

Table 3: Verification tolerance for EB2D tests.

## 12 Implementation Pitfalls (for the Reviewer)

1. **Section property choice.** EB2D bending in the $xz$-plane uses `props.Iz`. Using `props.Iy` is a silent bug in 2D (the field exists but has no role here).

2. $s$ **comes from** $\Delta x_z$**, not** $\Delta x_y$**.** The 2D problem lives in the $xz$-plane; $y$ is out-of-plane. Reusing a generic "$xy$" rotation formula will give wrong transformations for non-horizontal beams.

3. **Residual sign.** The element returns $\boldsymbol{r}_l = \boldsymbol{k}_l \boldsymbol{d}_l$, not $-\boldsymbol{k}_l \boldsymbol{d}_l$. The Newton solver assembles $\boldsymbol{R} = \boldsymbol{R}_{\text{int}} - \boldsymbol{F}_{\text{ext}}$ and solves $\boldsymbol{K}\,\delta = -\boldsymbol{R}$.

4. $\theta_y$ **sign under** $F_z$**.** A downward tip load yields a negative $\theta_y$ at the tip (test `test_eb2d_patch.cpp` explicitly checks this).

5. **Bending couplings.** The signs in $k_l$ entries connecting $u_z$ and $\theta_y$ at the same node are positive (e.g. $k_l(1,2) = +6EI/L^2$), while couplings between $u_z$ at one node and $\theta_y$ at the other can be positive or negative depending on which side (see the index map above).